



Challenges in Autonomous Vehicles: A Robotics Perspective

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Abstract—This paper studies autonomous vehicles as robotic systems, with a focus on their architectures, decision-making processes, and ethical challenges. First, the current levels of driving automation and the main components of autonomous driving systems are presented, including perception, planning, and control. By comparing two current design approaches, this paper discusses how system design affects safety, transparency, and ethical responsibility. The results show that ethical considerations should be an essential part of the development of autonomous vehicles, and not only a secondary concern.

Index Terms—Autonomous Vehicles, model architectures, neural networks, ethical concerns

I. INTRODUCTION

Robotics has evolved significantly over the past decades, expanding from controlled industrial environments to open, human-centered domains. Autonomous vehicles (AV) are one of the most visible results of this evolution, as they operate in real-world environments shared with human drivers, pedestrians, and cyclists. Unlike traditional robotic systems, autonomous vehicles must continuously perceive their surroundings, understand complex traffic situations, and make decisions in real time. These decisions can have direct consequences for human safety. Although advances in sensors, computing power, and artificial intelligence have greatly improved autonomous driving capabilities, technical performance alone is not sufficient. The behavior of autonomous vehicles is something that must be studied and taken into account before ambitiously wanting to reach the goal of complete automation. The topic of AVs has long been very discussed in the engineering in-

dustry, in positive ways, just like easier mobility, but also, negatively, where ethical considerations are discussed.

II. CURRENT SITUATION

As of today, there are six different types of levels of automation in automotive vehicles, defined by SAE [1]. These levels determine, in which way, a driver becomes, gradually, less and less active in the overall control of the vehicle.

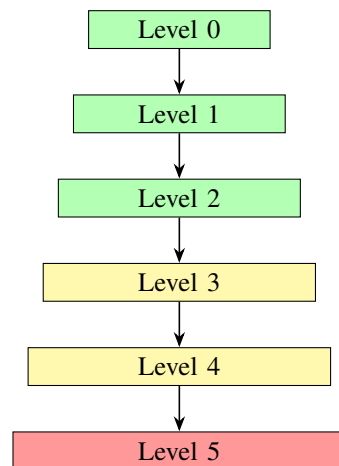


Fig. 1. Representation of SAE Levels for automation degree of cars

From levels 0-2, the degree of automation is only a support system for the human driver, providing assistance in braking, acceleration and steering. While helping out with avoiding accidents, by having tools, such as automatic braking, adaptive cruise control, or lane detection and

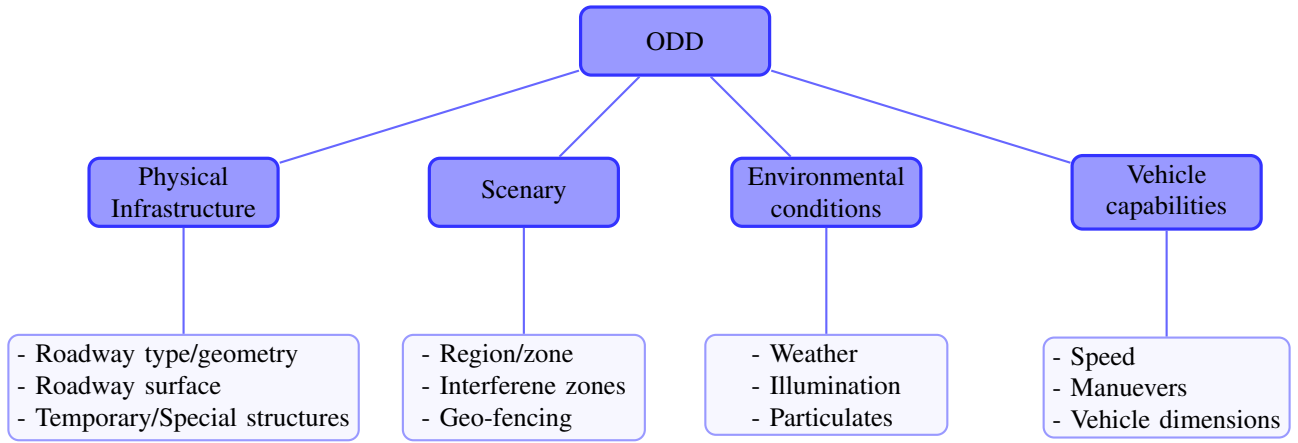


Fig. 2. Operational Design Domain (ODD) features

centering (LTA), the vehicle still depends on human intervention and supervision. Levels 3 and 4 make a significant advancement, from a technological point of view, since the autonomous system makes most or all of the decisions. The difference between these two levels is that, level 3 still requires a human driver, in case the system is not capable enough to do a specific task and switches off, whilst level 4 automation works efficiently and the system is only switched off by human order. The last level, still not reached, makes the vehicle completely driverless and free from any human attention. This makes it eligible for any type of driving scenario.

This method of classification ends up being quite general and does not necessarily surround all type of Automated driving systems (ADS). From empirical results [2], for a specific SAE level, some aspects are not very clear, such as, driving context, where the vehicle may have a different response. When and how the driver should intervene, based on the vehicles limitations (which could be undetermined) should come into consideration when developing a new taxonomy.

The Operational Design Domain (ODD) is an important concept in this context. The ODD can be defined as the conditions under which an ADS can be safely engaged, making it a widely discussed topic for several papers, which focus on creating a clear specification of the ODD. A commonly used approach is to design an ODD taxonomy accurately defining the conditions under which an ADS could operate. Some of the parameters, where the condition of an AV is analyzed, are represented in Fig.2. Other factors such as, traffic density and digital infrastructure also are taken into account. Overall, due to the non-redundancy and subjective nature of the ODD's taxonomy [3], there is no industry-wide standard used on an international level, leaving it up to individual manufacturers and suppliers to implement a consistent approach, that makes the AV better prepared for different environments.

The ODD taxonomy is more commonly used for level 3 or higher, where the ADS is responsible for the Dynamic Driving Task (DDT). Mercedes, the german car-manufacturer, has been developing an ADS, defined as

"Drive Pilot" of SAE level 3, and specify [4], after simulations and real test drives in various countries, an ODD that limits the engagement of the ADS, for safety purposes. Essentially, it defines boundaries (geo-fencing), where it is safe to use the feature. The feature will not be activated if the car is not in the pre-determined areas, which can only be "freeways". Even in those designated zones, if ODD conditions such as, the absence of stop signs, traffic lights or lack of lane markings, the system requests for fallback user.

A. Architectures

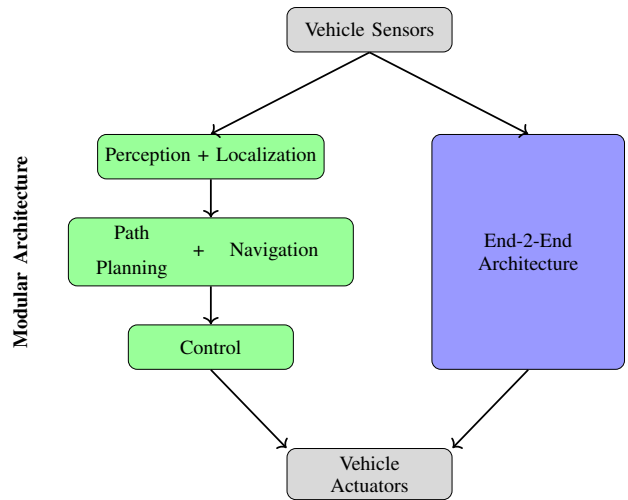


Fig. 3. Comparison between Modular and End-to-End Architectures

Inside each AVs ADS there are 2 main architectures, Fig.3, that are used in most cases: - Modular architecture; - E2E (end-to-end). The modular architecture is an older and more traditional approach, and uses a pipeline style method for specializing tasks, with each block depending on the previous block's data. There are distinct ways of separating each task [5], [6], e.g., depending on complexity and the hardware used. An E2E approach correlates all tasks into one, using Neural Networks, avoiding error accumulation caused by cascading and simplifying the process of learning. Inside a network, feature extraction from the sensors and cameras is the most important task, since it

helps the network model with training and work effectively [7]. Since the raw data perception and the output decision is all made in one link, the AV makes decisions on a quicker level, despite lacking on interpretability for the outcome decision [8].

B. Algorithms

Modular architectures have different algorithms for each task, while E2E has its own main algorithm. The tasks in the modular pipeline are the following:

- Perception and Localization

The most common type of sensors used in AVs are LIDAR's, GPS, cameras and radars. These have their high points, but also, some disadvantages, hence why in most ADS's sensor fusing is used. Current methods for sensor fusing are referred in [9], where Kalman filter's are the most used and traditional methods [10]. Sensor fusing, in Kalman filter's, is used by combining a predictive motion model with noisy sensor measurements, weighting each information source according to its uncertainty to produce an optimal state estimate. Other localization methods, such as, simultaneous localization and mapping (SLAM), YOLO (You only Look Once) and Vehicle-to-everything (V2X) are also quite regular discussed algorithms. Further details can be found in [11].

- Planning and Navigation

When the previous algorithms determine the positioning and localization of the AV, by having acquired and processed its surroundings, planning the next state is then possible. Based on predefined traffic rules (rule base decisions), vehicle states (MDP's) or other traffic participant's behavior (Game-theory), decision methods take place for outcomes, such as, maneuvering to the next lane to overtake a vehicle. The decision methodologies hold the highest degree of importance inside an architecture, as they dictate what the next move will be and for that reason we will discuss more about this later in the paper in sec.IV, relative to neural-network methods.

Typically, due to uncertainty and environmental complexity, the planning algorithm used updates the next state trajectory at a high frequency, based on the latest sensory information [12]. A* and Dijkstra are some of the more common search algorithms. Despite this, these algorithms are proven quite simple and unprepared for more dynamic responses [13]. Sample-based methods help address the types of path planning issues by generating candidate paths with random sampling. Rapidly-exploring Random Trees (RRT) have been demonstrated to overcome the disadvantage of high computational demand, since they only need to take in random sampling nodes to efficiently explore the environment, making them better prepared for high-dimensional problems [14] [15]. There are also optimization techniques that help optimize the goal for the path [16].

- Control

How much the car steers, brakes or accelerates, based on the input given by the decision, comes down to the control module. Simple module's used, mainly for longitudinal

control, are PID's [17] [18]. PID's are simple, but very efficient, which make them reliable for few constrained systems. Closed-loop PID's intuitively have better performance for harder tasks, but have a problem when it comes to tuning the controller [19]. As an alternative, MPC's predict the future behavior of a system based on a mathematical model. Thus, by predicting various behaviors, it optimizes the best one (J_{min}), based on predetermined constraints. It then applies the first control input in that optimal sequence.

$$J = \sum_{i=0}^N \underbrace{\|x_{k+i} - x_{ref,k+i}\|_Q^2}_{\text{tracking error}} + \underbrace{\|u_{k+i}\|_R^2}_{\text{control effort}} + \underbrace{\|\Delta u_{k+i}\|_S^2}_{\text{smoothness}}$$

The downside of this method is the conflict that it brings with the decision-method's input, x_{ref} . In simple terms, if the MPC does not find an optimal solution. Some approaches are [20], [21].

III. CASE STUDIES AND ANALYSIS

This section presents two case studies of level 4 autonomous taxi systems: Waymo Driver and Tesla's Robotaxi. These systems were selected because they represent different approaches to autonomous driving decision-making. Waymo adopts a modular and highly controlled architecture, operating within a restricted ODD, while Tesla follows a vision-based approach with increasing reliance on end-to-end learning methods [22].

A. Waymo Autonomous Driving System

Waymo driver uses LiDar, high resolution cameras and radar, which are strategically positioned to perceive the environment in 360 degrees. Before the vehicle starts operating (offline) in a new area, these sensors build a highly detailed map to collect dense data. When on the run, it continuously localizes itself, processing dynamic objects and changes to the initial mapping. Sensor fusion combines camera images and LiDar point clouds to improve robustness and accuracy. These sensors also feed data into a tracking system, which follow agents around the vehicle. Based on the current position and the velocities of the surrounding actors, LSTM models are used, to predict circumstances. Based on that prediction, Waymo uses a combination of rule-based decisions and planning algorithms, which have fallbacks in case of uncommon situations. For control methods, a hybrid feedback controller (MPC and PID's) is used. All this information can be seen in [23].

B. Tesla Robotaxi

The Tesla Robotaxi's system mainly relies on cameras and neural networks, without using LiDAR sensors. For now, this system is still not commercially available, as it's still undergoing vast improvements due to the dependence on collecting large amounts of real-world driving data from its vehicle fleet to improve system performance through data-driven learning. In this approach, 8 cameras around the vehicle generate 3D Vector Space, through neural networks, which are then trained to map sensor inputs directly to driving actions. In between those stages,

there are different neural network architectures used. Firs, RegNet and other residual networks work to extract a number of features of different scales and BiFPN combines those distinct features into multi-scale information. The vision system then converts the video-data into vector space, which then is used by both the explicit planner and neural network planner. [24] explains these two methods. HydraNet is the neural network architecture used, which combines the intermediate and output vector space features generated by the network itself. Thus, a trajectory distribution is created which is then fed into the explicit planning function. This function does the final judgment in terms of checking constraints and applying control to the actuators. The source for this information is at [25].

C. Comparative Analysis

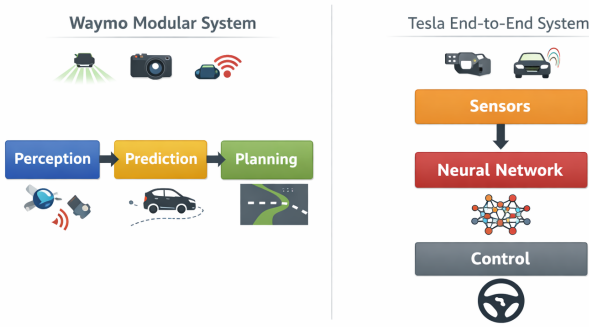


Fig. 4. Waymo's vs Tesla's ADS

The case studies presented in this section highlight two fundamentally different approaches to autonomous driving decision-making, each with distinct technical and ethical implications. Fig.4 provides a conceptual overview of these differences by contrasting Waymo's modular decision-making pipeline with Tesla's more end-to-end approach. Waymo's design improves system transparency and supports structured safety validation, as individual components can be inspected and tested independently. Such modular architectures are widely adopted in safety-critical systems because they facilitate verification and ethical accountability [22]. In addition, Waymo's ODD further reduces uncertainty by limiting deployment to carefully mapped and validated environments. In contrast, Tesla's approach reduces the separation between decision stages by mapping sensor data directly to control actions through end-to-end learning. While this strategy may improve scalability and adaptation to diverse driving scenarios, it significantly reduces explainability. As discussed in the literature, limited interpretability makes it more difficult to understand system behavior and to verify safety guarantees, which complicates ethical responsibility when unexpected behavior occurs [22], [26].

IV. DISCUSSION

A. Future Issues

On the basis of the two current model architectures, it is more likely that more investment is going to be made into end-to-end architectures. The reasoning behind this

is mainly because of a better tradeoff between price on hardware and efficiency. Particularly, taking into account Waymo's architecture, which base their perception systems on LiDAR, which come to an expense of roughly \$13,000, just for one AVs ADS. On another note, at this moment in time, likely due to Waymo's better sensor fusion [27], Tesla's RoboTaxi has a 58% higher risk of being involved in accidents [28]. For that reason, there does not fully exist a level 4 commercial product released by Tesla.

Due to the nature of neural networks, in order for the ADS to become more efficient, it depends on the size of the datasets, the richness of scenarios and also the distribution of data. Here efficiency means more stability and robustness in decision-making process. But a question surges, which is, would these factors still help the ADS predict and decide accurately in all types of unexpected circumstances?

To fix that problem, the system will need to increase its complexity [7]. This means that it will become increasingly more difficult to understand the behavior and decision-making of a model, which can make it lead into more road incidents, by not being possible to debug specific parts of the system. The ADS could also become overcautious and, for example, brake unnecessarily.

Besides all these points, more ethical concerns can be raised, such as, in the eventuality of a road accident, who's fault would it be. If the system's complexity is so high, how can investigations be carried out, since there is no valuable explanation.

B. Solutions

An idea to solve this black-box problem could be, focusing more on having a hybrid between modular and end-2-end architectures, combining tasks using neural-network frameworks, if there is improvements in overall performance. Each task must remain clearly defined and well understood, allowing targeted improvements to be made when necessary.

V. CONCLUSIONS

As seen in this survey, AVs are one of the most advanced and challenging applications of robotics, as they must operate safely in complex and unpredictable environments shared with humans. The study of current technological situation, in terms of level 4 architectures, showed how great advancements have been done in such a short amount of time. Both types of architectures have their pros, but also some disadvantages. Modular architectures provide better transparency and make system behavior easier to understand and verify, but do not have the same potential as end-to-end architectures, which will, in theory, get better the more they train. However, complexity will be inversely proportional with explainability. As ADS become more data-driven, understanding their decision-making processes becomes increasingly difficult. This creates important ethical challenges, especially when accidents occur and system behavior cannot be clearly deduced. A mix between both methods may offer a balanced solution, allowing both efficiency and interpretability in future autonomous driving systems.

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